

ActInf Livestream #037.1 ~ "Free Energy: A User's Guide"

Livestream | 1:55:01

all right hello and welcome to actinflab

live stream number 37.1 free energy a user's guide it's february 2nd 2022

22 22 20 days till the big one of course welcome to active inference lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at links here on this slide this is recorded in an archived live stream so please provide us with feedback so we can improve our work all backgrounds and perspectives are welcome and we'll be following good video etiquette for live stream maybe people can raise their hand visually so there's not a little jitsy blip until we figure it out or either way

go to activeinference.org if you want to learn more about participating or check the coda link here okay today in 37.1 the goal is to learn and discuss and also appreciate having stephen mann the first author here with us this paper free energy a user's guide and for this 37.1 we'll start with an introduction and then we'll just fire up some slides and uh here perhaps an opening and a contextualizing statement from stephen to be introduced last and then go into whatever questions that everyone else has brought and anyone watching live so i'm daniel i'm a researcher in california and i will pass it to stephen hello stephen i'm a practice-based phd uh researcher based in toronto and i'll pass it over to dean

morning everybody i'm dean i'm up here in calgary um i'm i'm somebody who does did some programming went from being an educator to a wayfinder and i find guides really an interesting topic to have conversations around i'll pass it down to blue hi i'm blue i'm a research consultant based out of new mexico and i will pass it to the first author steven

hi everyone i'm steven i am a philosophy phd uh currently a postdoc at the in the linguistics department of the university of surrey and a guest at the max planck institute for evolutionary anthropology in leipzig where i live all right maybe steven would you um or steven m would you like to just give a little context on the paper like how did this fit into your phd or research interests how did this specific collaboration come to be in the format of the paper and the way it is you know what i'm doing so i'm going to be able to do this uh first uh thanks for having me and thanks very much for last week's session of the dot zero i think you did a really excellent job at uh overview and i'm really looking forward to talking about it today um this paper came about because we uh myself and ross and michael the other authors had been in discussions uh with a few other authors around uh a symposium discussing active inference and the free energy principle and also uh predictive processing at the australian society for philosophy and psychology conference in 2018 uh we put together a small symposium um it went relatively well and um after a few more months of sort of bad passing ideas back and forth uh we approached michael weisberg the editor of biology and philosophy about putting together

uh topical collection sort of like a special issue but sort of um collected digitally and eventually

sort of after sort of pitching it to him and managing to get to talk to him in person um he said that he and the other editors were quite excited but what they wanted was uh an introduction from us that would explain the mathematics and the philosophical dialectics behind sort of at the core of active inference and the free energy principle in a way that was accessible for philosophers um and so we said sure we can do that and uh that was a little while ago we started soliciting out for papers we've now published most of the papers in that topical collection uh and since then uh at the same time sort of finalizing finishing off my phd and then moving to europe from australia where my phd was uh we were getting to grips with the real mathematical core of uh active inference and trying to figure out what was the necessary mathematics to explain the basics of the framework and then how to present that mathematics to uh a philosophical audience who has sort of some mathematical competence and the result is the current preprint so we think we're probably about one more set of revisions away from submitting for publication and it will eventually come out as the introduction to that uh topical collection so what i'm really interested in hearing from everyone else today uh this the same thing we're interested from hearing from the wider community which is the reason we released the preprint which is just comments and suggestions and any kind of corrections that can help improve the paper um so that its final form is sort of even better as good as we can make it

okay thanks for the awesome context steven s thank you yeah um we really enjoying this paper and one thing i'll be interesting to to think about even though you you sort of aimed it at maybe a philosophy audience is some of the aspects of it at this nascent kind of foundational level could help in an applied context that may be uh beyond the modeling uh so we're kind of interested in some of that so that might be something we might tip into even if it's not in the remit of this paper's direct publication i think it would be really helpful in the broader sort of journey of using active inference in the world

awesome okay um blue or dean would you like to start with a question or we can return to one of the points that were already kind of brought up i'm good for now um okay so maybe just one starting point was just what backgrounds like how did you go from thinking about who you wanted to make the paper accessible to to writing it a certain way and then you mentioned it was for philosophers so what is the impact in this community that you'd like to how would you like to update their generative models uh first and foremost we would like to fix in their mind the relationship between uh variational inference and bayesian inference so i know that's something you've uh talked about before it's something that predates the free energy principle and active inference as an explanatory framework for cognitive science and biology and it's something which as far as i can tell is sort of the first step in understanding what's going on with that mathematics um so that's the sort of the key thing we want philosophers to sort of have in their minds also because almost all philosophers especially philosophers of cognitive science but uh but also philosophers of biology will be familiar with bayesian inference at least to some extent so by pointing out that what active inference is is something sort of a step to the right with bayesian inference something that they can start to understand by analogy with that we hope it will stop them from sort of getting scared of all the complicated mathematics that you can read in a lot of the papers um so that's on the math side on the philosophy

side a lot of the stuff in section four of the paper was our attempt to really strip back uh a lot of the claims that were being made but also sort of categorize the claims that were being made one thing that i noticed in reading the literature and attempting to understand it was that different kinds of claims were being conflated and there was jumping around from claims in different domains mathematical claims that suddenly uh leverage to justify an empirical claim and then suddenly move to an extremely general claim about all of cognitive science or all of biology in general so that's the second thing we want the philosophers to get that it is possible to disentangle these claims despite the fact that they are habitually sort of conflated or not made distinct enough in the rest of the literature all right one follow-up and then i see steven and dean um you mentioned that bayesian inference may be more familiar especially because it's used in all kinds of modern statistical descriptive approaches as well as generative just to hear it in your own words where do you think variational inference is more familiar so who does it represent who's more familiar with variational methods or who would be familiar and then what are the key differences sure so my understanding is that uh physicists statisticians and machine learning practitioners uh may well have come across variational inference certainly will have had a much better chance of having come across variational inference than philosophers at least as of 2022. um what i understand variational inference to be is a certain kind of technique for choosing a probability distribution over unobserved uh states that started uh in physics as i understand it with fineman although i put a little footnote in the uh preprint pointing out that i haven't been able to translate the stuff in that fineman book to the later stuff especially stuff that um david mckay talks about in his textbook on information theory which includes some issues in variational inference but as i understand it started in physics uh then was adopted into sort of statistics more broadly as a means of approximating bayesian inference in cases where exact bayesian inference is too difficult to carry out too computationally difficult uh and then moved from statistics into machine learning where it's possible to actually build machine learning algorithms or systems that can perform variational inference and then from there it ended up in in the active inference framework all right thank you so dean first steven thanks for joining us i really appreciate it um here's a question for you now that now that uh uh an examination of the situation and the situation as you stated was kind of um math and um philosophy have now i've got a relationship and we're going to pop up out pops a guidebook or a primer when the when the pump is primed what do you see as being the effect of that that relationship between the sort of philosophical audience and the and the math that was used to get us to this place relationship between the philosophical audience and the math that was used to get us to this place i mean first and foremost i want the philosophical audience to gain a certain kind of facility with the math i think one of you i'm not sure who of you it was but in the point zero put it extremely well by saying it's like learning how the horse moves in chess i think it might have been daniel but i don't want to misattribute it to you giving the philosophical audience the the basic understanding of the most basic maths will give them that first kind of tool which they can then go on if they so wish to

build up and learn more of the moves learn more of the complex maths if they want to then go on and interrogate the rest of that literature which is more complex we've tried to sort of forego too much critical analysis in the paper there is some in there um we've tried to forego that a little bit just in order to get the understanding of that maths across okay

i can follow up there um what does it matter for the philosophers so what are they in your sense or their sense what will knowing the difference between bayesian and variational help them evaluate or what will seeing it structured this way help them evaluate

i think there's a few so i hope other people also can think of philosophical questions that knowing the difference between the two would help you change your mind on

yeah i mean the simple answer is just it will help them understand what fristen is claiming and by fristen of course i'm using him as just an umbrella for anybody who's writing uh positively in the active inference tradition attempting to demonstrate that the active inference framework can provide or inform certain process theories that uh explain cognitive or biological systems philosophers need to know the difference between bayesian inference and variational inference because they need to know uh what it is that fristen is claiming can be attributed to systems like this okay blue on that

so i have a question and um like maybe i missed it and i was looking i just did like a quick control up on the paper just to make sure that i didn't miss it and i'm not like misinterpreting but um in your explanation of um the generative model like i just wonder why you didn't choose to go into the recognition model and i think that like what you discussed at length like p's and q's um why did you leave that out or do you think it's not significant or or did i miss it and did you talk about it at length so by the recognition model i understand that to be a sort of alternative way of doing inference that uh an agent could employ so i'm just i'm saying what i understand by it so you can correct me before i try and answer the question uh i understand it to be yeah an alternative to generative inference is just to employ a kind of recognition model which simply takes in data and immediately produces some kind of statistical result from that data it doesn't need to uh simulate or generate its own data and then compare that to what it sees that would be the generative way instead it just applies some kind of say mathematical statistical function to the data

and and then gets a kind of result which describes what it hasn't observed is that what you meant by recognition model so no so i'm referring to um the paper the tale of two densities the 2019 paper by um bramstad al and um you know in that paper they they refer to um the recognition model of the likelihood model as the q right the inverse of the likelihood model it's the mapping from the consequences to the hidden causes and so like you you got really into the maths but and discussed the the um generative model but just didn't um like acknowledge this this recognition model and i just was curious as to why okay so you just mean the the distribution q

uh by by the recognition model um um am i using a different q i haven't read a tale of two densities

i should say so uh yeah the recognition model is on the it's um in the uh kind of namespace of expectation maximization modeling the recognition model is like the sensory update and then the generative

part was like emphasized i hope that's not too simplified or what two densities were and then the outbound was more like associated with the generativity and that's also um that's why it's important to go from the frequentist to the bayesian at all because with bayesian generative modeling like the model can be run both ways though it's used with different adjectives that's like variational auto encoder that's um all these methods where it's not just descriptive modeling on a data set like principal component analysis or multiple regression it doesn't just describe then it can be used although those can be used or be thought of as bayesian like and be used in a generative way as well but that's the part that people will have less empirical experience

that's seeing but let's think about this and then return to it next week um yeah okay so stephen s just following on um from the bayesian um which is often used in the embodied context people think about embodying and multi-sensory integration and that type of work and that field is a bit challenged by moving over to this variational machine learning paradigm because it's highly mathematical but what's interesting with active inference it sort of comes back into that being grounded in embodied action because that gives the preferences somewhere to be anchored so how do you see um this going back into embodied cognition as a way for those preferences to be grounded

that's a difficult one uh for me to answer in part because i'm not really part of the tradition that emphasizes embodied and inactive uh approaches to cognition um the way i was taught it they were sort of alternatives to the representationalist tradition and i was much more in the representationalist tradition so there has been some talk of whether and how free energy approaches support one or the other of those two

traditions uh and to some extent because i'm already sympathetic to the representationist tradition i'm already leaning towards the idea that uh free energy approach will support that you know as opposed to an embodied or an inactive kind of approach which is sort of a shame because like all of those kinds of binaries it's probably somewhat a false dichotomy uh and it just means that you know my direct my attention has been directed elsewhere than your attention in looking at these issues um and so yeah i don't have enough of value to add unfortunately

i think it yes stephen one one second but um our previous paper 36 was like a taxonomy of perspectives on the representation question and fep so i think we're all very primed to just take a totally open stance and is is every possible stance for every person to focus on it's just kind of not within our time and attention capacity so i think it just speaks to um like a diverse field where of course people are focusing on chemistry of this element of that element it won't be exactly that split but in the broadening field of like active inference and fep there's going to be people who are focused more on one question or other so that's kind of cool to like know where we um have similar regimes of attention or not stephen yeah i'm agreeing with daniel and also actually we found this really really helpful from an embodied perspective because the way that you separated or you gave the probabilities and preferences sort of working together was um it was kind of like a deflated or a more nascent form of representation and that gives quite nicely with potential ways that inactive approaches are also potentially there because of the nature of preferences so you start to have a way where they can both live together

because we i don't know we were talking about this the other day weren't we daniel that that it wasn't so clear how these two can live together and often when people who are from philosopher quote the free energy principle they talk about early papers like 2000 you know maybe 10 years ago even and that um is often then a false

argument that gets painted so now we've almost got a new paper which can kind of help ground those philosophers so i'm wondering what your thought is in terms of whether you're trying to help people use the most latest form of free energy principle when they were talking about um philosophy and to stop some of the conflation of just people criticizing models which are now out of date um if that was part of your rationale for the way this paper was created yeah we absolutely wanted to direct people's attention to what we thought of as the real issues in a way the the appropriate issues um i mean we reckon that a critical assessment that is not well informed by the actual workings of the theory is not going to be helpful because you might accidentally hit on a critique that works but only accidentally if you're going to criticize a framework you better at least understand what's going on um my worry is that um it will become out of date as things change but there's not a lot you can do about that i mean it took us quite a long time to get this first draft together to get this preprint together and sort of hoping that things haven't changed so much that it's out of date by the time it comes out uh i mean one way to forestall that is to just be as basic as possible so that you're really capturing the core that can't possibly you know change um but otherwise your best chance is just to publish what you've got at a particular time and keep your eye on how things change and update as you go thanks yes dean steven if i use a lot of metaphors so if you walked into a singles bar and one of the singles was a philosopher and the other single was a mathematician and you had to guess now going through the process of building that guy right like you're not match.com but right you have to make a guess as to whether that long-term relationship they walk out hand in hand or they're completely repulsed by one another i know it always depends but you're talking about updating the guide i'm kind of want to go back one previous step what keeps them together what pulls them apart yeah so how do you write what is essentially a philosophy paper but include mathematics in a way that does justice to both is is roughly how i'm hearing that question a long-term relationship right like they still gotta have some compatibility and that's the ability right so its point was up to now it's been i don't know if i even want to go to the bar so that's that's what i'm asking you now based on what you've what you've journeyed into right because not everybody i know i understand i appreciate the idea of the guidebook because not everybody goes into situations freely right sometimes you get dropped into situations and it's a little bit more intense but what causes people to go freely into that kind of uh relationship and sustain it over a long enough period that it does make sense it doesn't remain kind of a superficial thing or just a concept but it actually becomes realized i was curious what your thoughts on that are

um

i think that

you know trying hard not to repeat myself too much uh the way i the way i try and think about

that question is to try and think about what has prevented philosophers from really accessing the mathematical parts of active inference so far and that's just from sort of communicating with philosophers and talking to them and hearing their dissatisfaction with it and it was always just that there was there was too much of it too fast and so that the sort of the only approach we could see that could remedy that was to go as slow as possible in a way uh and so spend sort of have a much heavier ratio of words to equations in a way and go into much, much more depth in explaining each equation as it comes along rather than throwing all in at once and then decomposing it.

I mean, one of the earlier versions of the draft, we started with the main equations and then decomposed them.

In the current draft that you've read, we start with the components and then build up to the main equation.

That was a little bit risky, I thought, because I was worried that people would feel we were holding the main event back for too long.

But so far, it seems to be working, I'll say.

I'd be interested to hear your thoughts on that approach, whether you think that meshes together well enough.

Yeah, no, that's one of the things.

And again, I don't think mathematicians are trying to be pushy.

But if you don't have that context, it can feel pushy.

And so how long you're willing to leave something open before you drop the big question is sometimes easier for somebody then to size the situation up and contextualize it.

So in this delivery, I don't know previous versions.

But in this delivery, in that respect, I actually think it makes it less intimidating.

But that's just my personal opinion.

Sorry, Stephen, I know you wanted to say something.

Stephen.

Yeah, Stephen.

Yeah, just carrying on from what Dean was saying as well is, you looked at the inference model almost prior to having a Markov blanket, you know, and I thought that was quite useful.

And one thing that that made me think as I was reading it, and I'd be interested in your thoughts is, it helps with this idea of really thinking about where, how are we going to create the blanket when does the blanket come into play and what assumptions are there about the blanket in some ways um it's easy to pick up a pre-built idea of what blankets are and then model off that whereas it's like okay well let's question that and that actually could be useful at your a more heuristic level you know for other fields of practice so i'd be interested in your thoughts about what do you think your motivation for that and also maybe the sort of surprising or unsurprising outcomes of not going straight for the blanket

well one of the big revelations for me in researching this paper was that you can talk about variational inference and you can talk about minimizing expected free energy without talking about markov blankets i had sort of come into this with uh just sort of a huge deluge of literature from the active inference tradition lots and lots of like key words or what could more uh uncharitably be called buzzwords like markov blanket and non-equilibrium steady states and things like this and it was a concern that i would first have to understand statistical physics and fluid dynamics before i could understand the claims that were being made about inference but looking back into the history and you realize that first of all variational inference is a precursor to active inference so you can think about variational inference in its own little bubble as we have it in its own little section and then the sort of next step which is novel and is quite excitingly novel um which is minimizing expected free energy can also be construed just in terms of decision theory you don't as far as i can see you don't have to think or talk about markov blankets at all to talk about decision theory as minimizing expected free energy so the reason why we separated them was because they're separable the reason why we put those two first variational inference and expected minimizing expected free energy is because i think they are simpler and easy to grasp and we were trying to start from the simple and build up to the more complex it's i think it's definitely true that the most sort of intriguing and possibly the most sort of difficult part of understanding this whole system which friston is building is understanding the things to do with markov blankets in real like physical situations um but what i'm sort of interested in now to give you a preview of what i'm like thinking about next the more the thing which i'm more optimistic about being able to make headway on is investigating minimizing expected free energy as a decision theory that sort of gets the middle ground between it's not just something that prefigured active inference as variational inference did it's not something that seems to require a very large amount of mathematical understanding as the markov blanket stuff does seem to to me it's something which is claimed to be a certain way of approaching decision theory decision theory is sort of a philosophical sort of technical term or statistical term so it's that nice middle ground between novel but understandable um which is why the yeah the thing that i'm sort of most interested in investigating next is uh the expected free energy approach and again i don't think you need to talk about markov blankets to talk about that right okay thanks for sharing that one thought um on where the markov blanket comes in so you said that variational inference and the minimization of expected free energy in the context of a paper that was structurally laid out with first perception and then introducing action into the loop so it kind of fits this perception of case is like action affordances are not possible and so that's the special case the more general case is when action can come into play that whole realm of action theory decision theory game theory cybernetics signal processing none of that the way is the more general case is the more general case is the more general case is the more general case i'd like to maybe add that there may be a weak sense requirement like in the sense that on a bayesian graph or using bayesian methods using pearl 1988 version of the markov blanket there are markov blankets in the machinery of these bayesian

methods

because there are variables that are isolated or conditioned on each other uh so that's like the trivial Markov blanket

the non-contentious one that people are not spending a lot of time and attention on

that um and then part of the unknown or at least less clear is the action and perception interpretation

or concordance that Fristin has introduced, kind of taking the unitary Markov blanket concept

and splitting it into like an incoming sense state dependency and outgoing action state. And then

none of that gives a clear action perception topology, which is why we've seen like around

the clock and then with the ones like action and sense are connected and backwards arrows. So

there's the part about the partitioning of the Markov blanket and then taking something that's

kind of a trivial consequence of statistical models and then tying it to potentially realism

in the strong cases, like people saying Markov blankets as defining things. And then even just

within the formalism, how do you go from, how do you ramp up the meaning of this Markov or does it

need to even be that way? Which is kind of related to this question, which I'll ask and then anyone can

give a thought, like you talked about necessary and sufficient. So is this some of the necessary and

sufficient pieces of FEP? What is, what is necessary? Like what's the kernel that everything else can

continue to elaborate around? What section and page number please?

Well, you'll notice that we were cheeky in defining three distinct free energy principles, probably

none of which would be, you know, assented to by the actual proponents of the free energy principle.

Yeah, exactly. Could you go through and just describe in a few like sentences or just what, what is each one of these, the three free energy principles, inference, action, selection, what did they mean to you?

Right. First one, free energy principle for inference as, you know, as we've defined it is

when updating your, sorry, my cat is biting me now.

Sorry, I'll hold him up here so it doesn't bite me.

Um, well, the short version is choose Q by minimizing F . So choose your beliefs about hidden states

and the free energy principle for action. So it's a principle in the sense that philosophers would call

a normative principle. It's a command for how to act. In this case, the specific kind of action that is

how to choose your beliefs.

Okay.

Okay.

And, uh, free energy principle for action, also a normative principle telling you how to act.

And this one is, uh, in our, again, in our terminology, choose Z by minimizing G .

So choose your action by minimizing its expected free energy.

And this is all for anybody who's watching this, who hasn't looked at the paper, we've

tried to sort of describe, um, in very slow, but understandable way, what, what these different

sort of letters are standing for. Uh, let me get the definitions for a discrete case.

Okay. So we have perceptual inference, Bayesian expectation maximization, like choose Q , the

member of the variational family distribution by minimizing this snapshot F based upon your priors,

basically, and new information coming in. And then that takes on a normative interpretation.

It's kind of like saying, if you were doing linear regression, you would do the L2 norm

least squares. If you're doing this type of regression, then do this kind of inference.

The second one is also normative, but it's action theoretic because you introduced action

and then that introduced the elaboration on F to the G function.

Um, maybe Steven S. How would you say that this movement into the future first matters for maybe not necessarily how a philosopher, but how, um, any other relevant area might see this in an applied context? Like what does the F to G movement have to do?

Steven S. Well, I think it's, I think what's useful is that it's like, it's almost taking what's evolved out with a Markov blanket with the expected free energy and bring it back down into a more a form, which isn't only a form, which is involved in the modeling out of particular systems at low dimensionality. It's coming in and say, okay, this is a broader principle that can be talked about.

Steven S. And I think that then ties into some of the stuff the active inference labs trying to look at is how can active inference itself be used as a way to think of organizing and structuring our lives, like George Parekh might have thought about it. And, um, how can it be, um, used as a, as a heuristic, um, in, in, in areas like coaching psychology in areas where you're looking at action orientated decision support in complex evolving ways, you know? Um, so you can't, you can't model it in the, in the traditional sense of a Markov because there's nowhere near the dimensionality can be approached. So what's useful is they say, okay, look, we've got these general ideas. We've got a route to it, but that this, the general deflated forms of that actually then because they're not so high order and they're not so, uh, nailed down to one idea of a Markov blanket maybe has a more universal application when someone's trying to look at complex contexts and how to act in the, in the world.

So I don't know what your thoughts on that, but that that's, that's what I'm very interested in is this idea between sort of diagnostic kind of syndrome based psychology and how do we look at actually what's going on in a context, look at the, the symptoms, look at the context unfolding and work on modeling using our own apparatus as dialogical sort of collaborative human beings.

Stephen, uh, uh, Dean first, and then Stephen M if you want to give a thought on that. And we're just going to talk about this inference and action piece, because I think there's a few more pieces to draw out and then we'll get to the third selection, but Dean go for it. Yeah. So back to my strange twisted metaphor. So now all of a sudden the philosopher walks into the bar and instead of one, one math showing up, we'll call it the geometry, which is much more amenable to the model in terms of how a philosopher might walk into a space and see the geometry first. Uh, all of a sudden there's two maths. There's a statistical math that's, that's showed up at the same time. Are they twins? What's going on here? And I think that's where the guide is kind of handy because if you first walk into the space, I'm, I'm not arguing that a Markov blanket exists. I think relationship matters, but I think what most people that walk into that bar don't realize is that you're walking in on two different maths.

And they're both showing up at the same time when you were expecting one, you were expecting to be able to work with a model, which is, as I said, it's far more proximal to the geometry math than to the

statistical one. And so I think that again, when we're talking about guides, um, I don't have to, I can put my hand in the ocean and figure out what the current is. And I can look up at the stars and figure out where I am in that invariant space. I don't necessarily have to have a laptop with the statistical distribution. I mean, that might help, but that's not a tool that I need it firsthand. Whereas I think the modeling part, which is what you were, were trying to bring up with this guide is at hand. And so it's the availability piece that I think philosophers are trying to, are trying to, what, what, what, what did I walk into here? That's essentially, what am I walking into? And what is the betweenness of those two types of math that they're probably, if they're like me, but I'm not a philosopher, they're asking when they're, when they're being introduced to this. So I'm just curious, again, if I'm, if I'm going way off on a tangent, or if we're introducing something, we actually know what we're being introduced to. And that's the part where Steven used talked about slowing down, because I really agree. It's not a breaking down. It's an actual slow down. Nothing wrong with that. Whoa.

I mean, I could try and approach an answer by sort of extending your metaphor, if you'll give me license to do so.

Absolutely.

Imagine you're going to the bar, and there's dozens, or even hundreds of maths or different components of maths or different approaches to maths. And what you'd really like is some way of pairing that down to something that is approachable. You know that only a few of these will be approachable, but you don't know which ahead of time. I'd like to think that you could use what we've written as at least a first step to pairing some of that away. So what we've left out, essentially, is the stuff that you could pair away and say, I can't approach that yet. What I know I can approach is these first few things. And what I like to think is that we've done okay at getting at least some of those first few things in there. Not too much, not too little. It's really trying to find that gold luck zone. And those will be the ones that are approachable, even approachable for a philosopher who's met very little maths before. I know I owe Steven S. a response to your earlier question as well. But if you want to go ahead first, Steven or Daniel, you'd say what? Yeah, Steven S. go for it. Yeah, I was just saying maybe bring up slide 13 if you're able to, Daniel, because that might be a useful one. Yep. Just as it has.

I have it. Yeah.

Cool.

That's got a little bit there about the penalty for overfitting, penalty for failing to explain the data. And that has 13.

Oh, is it? Oh, wait a second. Maybe I'm looking at the numbers at the bottom. Okay.

But on page 23 or slide 35 here.

Yeah. Yeah. Okay. Yeah. Slide 21.

Yes. Yeah. This definitely juxtaposes the F and the G. It's like a Pokemon evolution,

but it includes those different pieces about bringing in action with the uncertainty on the consequences of action and the fundamental and the ability to impact it. But that was actually what I had kind of written down. So Steven made the point about how action is a lower dimensional space. And that's like making an algorithm tractable, like conditioning inference on action, which is something that's very different structurally than a lot of other control theory, where the target of inference is a policy selection. Here, the affordances play a little bit of an upstream role and they make the downstream calculation conditioned on action. So that speaks to a different embeddedness

of action in active inference. And then there's a few other things, but first, Steven M, what was the answer that you owed the other Steven? Yeah. I mean, I was really just picking up on the point that Steven made about the difficulty of accurately modeling. And we talk about this a lot in philosophy of science. I mean, in the philosophy of any of the special sciences, but in the sort of philosophy of science more broadly, one of the biggest topics is modeling. And one of the biggest questions is, how do you trade off between getting a model, which is a perfectly accurate representation of the thing that you are modeling versus something that is tractable and that you can handle? And the analogy is always made with maps. That's sort of the first analogy. You can't have a map that is an exact replica of its territory because you wouldn't be able to use it to navigate. It could not provide you with anything that was more helpful than the territory itself. So we have some kind of practical knowledge of how to make maps. We know what to leave out. We know how to condense it. We should at least,

you know, shrink it usually if we're creating a map from a territory. What we don't really have in a very general sense across all of science is a set of principles for how, given a specific situation that is too complex to produce a perfect representation of, what ought you leave out and what ought you keep in? And how should you transform that space into a representation of that space in order to produce a representation that is a useful approximation of it? So the question coming, you know, back to the formalism would be, how do you choose a kind of a Q distribution, your approximate distribution, the thing that you're going to harbour in your brain and believe that throws out enough of the useless stuff of the real world, but keeps in enough of the stuff that you really need to keep in, in the world. This is sort of the, one of the big questions in philosophy of science and I guess science proper at like the early 21st century. Awesome answer. Thank you, Dean.

And then Blue.

Yeah, just real quick. So Daniel, when Daniel talked in the point one about the third P, but he called it the P-H-I-T-N-E-S-S, the fitness, that's what basically you're describing. How do we take that that variance field and collapse it down to something more tractable as you said? And so at some point later on, either later on today or in our next live stream, I think that's what the active inference model, which is not a map, and it has probably two parts. It has a frame, but it also has a, it also has a filtering effect. Maybe we can talk a little bit about that because that's, that's what the fitness is really

about. So I'm looking forward to sort of diving a little bit deeper on that because it's not just a back and forth across and, and reaching edges. It's actually going deeper into it and then returning from. So yeah, I'm kind of looking, looking forward to that from a mathematical standpoint as well. Thanks Dean. Blue.

So that's a perfect like segue fitness into what I was going to bring up. And so, Stephen, something that you mentioned earlier is like your attempt to capture the essence of the FEP and the things that will not change and kind of build slowly, incrementally. And so if we have a theory that does not change, essentially like does not evolve, it will die. Like it will become extinct because we're constantly learning new things. And so, I mean, I think, um, I, I, I like from a counterpoint think that it's kind of critical to capture the evolution of the FEP over time because it really changes the, the potential applications and, um, ways that, that people interpret the, the principle and the theory of active inference also. And so what, like if it doesn't change, I mean, will it not ever change or is it subject to change or just what are your thoughts on that? Like, like, is it totally inflexible? These, these points that you've attempted to lay out?

No, I mean, yeah, that general point is very well taken indeed. Um, I, I guess I think that, uh, by trying to capture something that we thought wouldn't change, I think we meant, or I should have said, we're trying to capture the parts of it that the rest of the framework will continue to depend on, um, for sort of as long as possible. We tried to keep the most stable parts of it without necessarily implying that active inference could survive without these mathematical components, but just that, uh, the change to the theory or the change to the approach would have to be so radical once you'd got rid of these core mathematical components that someone could legitimately ask, is it still the same theory? So that's roughly what I would mean by essential such that if, for example, tomorrow, Fristin were to say, we don't need to talk about variational inference anymore, or we don't need to talk about minimizing expected free energy anymore. Someone could legitimately ask, okay, is he now talking about a new theory that's no longer active inference? Is it so different because he's thrown out that sort of fundamental part that he's now actually proposing something completely different? Um, I think that someone could legitimately ask that question. I think that a reasonable person could say that he's now talking about a different theory because I think that those mathematical parts are so crucial to it as opposed to some of the other parts which seem some of the more complex parts seem more dispensable. You could throw out some of the more complex things, keep these more basic things, and you would still more likely be having the same theory.

Yeah, Blue?

So just to respond quickly, like, so for me that's kind of an arbitrary line in the sand. Um, and like one thing we talked about back and forth a lot is this concept of ergodicity and like an ergodic system versus like a non-equilibrium steady state density, locally ergodic, like what does it mean? And is the, if we throw out over density and eliminate this requirement, like is the theory still the same theory? And so like, that's one point, but I mean, there are many points, um, in like the minimization of free energy and, and so on and so forth. And so as we add on to the theory, we are changing the theory all the time.

Like as new papers come out, like the recognition model versus the generative model and you know, Bayesian mechanics for stationary processes was kind of really highlighting a couple key, um, points, but, but as the theory goes and changes, it's like at, at any point we could say, well, that's not the same theory. And so if you're, if you're, so the theory is not allowed to evolve over time, like it's just this arbitrary point in the sense that, that you're, you don't want to cross like the minimization of free energy. What if we decided that the minimization of free energy means something totally different? Like even in this paper, I mean, you had in a footnote, um, about the complexity versus accuracy, but, but to refer to free energy as inaccuracy is like, I've not heard of it described that way before. And so for me, I mean, it is this complexity and accuracy, and I understand like the, um, the way that you framed it, but it's a completely like novel way to talk about free energy. Like I always hear about as here, here it's in framed in terms of like uncertainty or surprise, but not inaccuracy. Like that was completely like caught, took me off guard actually. So, I mean, as the theory evolves, like, I don't know, it's just, what are the, what's the irreducible component of the FBP? Like does such a thing exist or is it just turtles the whole way down? Yeah. You're asking me the really hard questions. I like this because it's tough. It's really tough to think it's honestly, yeah, it's tough to think and answer that. And to, to ensure that I'm answering sort of honestly as to what I really believe, as opposed to just trying to sort of dismiss what you say, uh, you know, I, I'm not trying to dismiss what you say. I, I do think that the line is blurry, so it's not clear and distinct. I don't think that it's arbitrary. I might be wrong in that, but I, I don't believe that it's arbitrary. Um, let me just leave to the side for a moment, the, the description of free energy as inaccuracy. Uh, I'll come back to that because I'm interested in picking up what, yeah, what you think about the different interpretations of, of the function F .

Um, but I'll try and go back to sort of the idea of erbificity. Uh, yeah, there's a sense in which things are still being worked out. And one of the things we put in the paper, um, is something that we were urged to, uh, make clear, which is that active inference is a work in progress. Even the sort of mathematical underpinnings are still being fleshed out in a sense or still being put together in a way that helps connect the different parts of the theory. Okay. That's okay. That, that in fact, was a little bit difficult for me to swallow at first, because I felt that if that's true, then it doesn't legitimate all of the strong claims that are being sort of made, but all of our complaints about that are in that fourth section of the paper. Um, so we tried to make clear what we think is sort of justified and not justified on that basis, but it's still true to say that it's fair to have a framework and to, you know, to expend time trying to put together the mathematical underpinnings of this framework, uh, even though they don't yet exist, you know, we don't have to, we shouldn't just throw out the theory or disparage it just because the full maths hasn't been worked out yet. I agree. We should allow it to change. We should allow it to evolve. We should allow it to be, um, applied to different things and see what works and what doesn't. With all that being said, I, I don't think in this particular case, maybe it's true of other theories, but for active inference, I don't think that it's true that there are things that you could remove

and still reasonably say, yeah, this is still active inference. I think, I think a theory could not be that amorphous and still be true sort of treated seriously as, uh, as a way of pursuing science. I think that you would have to say if you threw out minimizing variational free energy or you threw out minimizing the expected free energy or some other of the core components, I think you would have to say if you're being reasonable, again, the word reasonable is doing a lot of work in this paragraph that I'm building up to. I think you would have to say, okay, it's a different theory. I think you'd have to say, okay, we were wrong to pursue the minimization of variational free inference, the variational free energy as a principle for understanding the behavior of biology and cognition. We are wrong to pursue that. We're going to pursue something else instead. And so you couldn't just say, oh, it's still the same theory, but we've changed this core component of it. I don't think that line is arbitrary. I think there are some things that if you change, you have to sort of admit that you're wrong about that approach and go onto something else. That's sort of my piece. I've talked for a really long time. I do want to pick up the idea of free energy as a function that measures inaccuracy, but I don't want to talk for like a huge amount of time. So maybe I should put a pin in that and we could potentially return to it later. How about first free energy selection, Stephen M finished that three, then as the inaccuracy, then I have a point.

Stephen M finished that three. Right. Great. Yeah. I forgot we were only two thirds of the way through our descriptions of the different free energy principles. I'm sorry. So free energy principle for selection. Let me try and pull it up here so I can just read it directly off the page. What we have it as is any system that survives long enough will act so as to appear to be minimizing F . One of the issues here is that we've left out G , but for the sake of sort of focusing our attention, we can just talk about F .

Okay. And then how about the FEP as, or free energy as inaccuracy interpretation?

Great. Yeah. So this is, this is something that I was hoping would be again, a way to present it to philosophers so that it would be easier for them to grasp what was going on. I wanted them to understand why do we care about this function F ? Why do we care about variational free energy? Well, suppose that I tell you that in order to do inference, what you ought to do is minimize variational free energy. And then you could very well say, why, why should I care about minimizing this thing? And then if I tell you, oh, it's the measure of the cost of inaccuracy in belief, then you immediately say, okay, I understand why I would want to minimize that because I want to maximize my accuracy. So very simply put, it's a way of explaining why we should care about F . It's a measure of inaccuracy. I think that that interpretation holds up just to the extent that if you look at the two components, which can be sort of thought of as, you know, a cost penalty for overfitting cost penalty for being inaccurate. There's a part of that, which is inaccuracy. It's not the whole story, of course, because the other part is to do with complexity, not just to do with accuracy. So what I think we've done by describing free energy as a measure of inaccuracy or the measure of the cost of inaccuracy is trading off that sort of ability to enter into understanding the maths for, you know, a perfectly clear, a mathematically very precise way of describing it. But I think, I definitely think that it should be thought of as containing

a term which is a cost of inaccuracy. So we talked in the dot zero about it's kind of like finding the shape from the data and the fineness of the mesh and inaccuracy in the real world kind of has elements of both or at least we're used to thinking about complex decision trade-offs. Like you could make a bad decision because it took too long, but then you came to the right outcome in the scheme of thinking bigger than just one p-value. And so it's kind of opening up even this very kernel piece of the puzzle to the whole question of how specific and how many subgroups and always leaving space for not overfitting, given this whole map territory Borges discussion. Stephen S.

And would you say that the overfitting inaccuracy, you know, is more around when you talk about adaptive, active inference where there's a way to adaptively look and have an idea of beliefs in some way.

And that underneath that, there could be a kind of a more abductive, swarming kind of entropic kind of process, which is sitting in the background. But in terms of how we normally explicitly think about meaning, we would think about it more in terms of what can be understood as beliefs. So I'd just be interested how that, how you see that from a philosophical point of view.

Well, one of the things that's still unclear to me about the overfitting part of F is that it seems that it can only be interpreted as sort of a cost of complexity, if what you already believe is simple. Because what it's really measuring is how far you are straying from what you already believe. So that there's kind of an implication there, that everybody always has simple beliefs in the first place. And that therefore, they ought not stray from those simple beliefs. One of the things that I'm trying to think about now is what happens if you are starting with sort of more complex beliefs? Maybe that's not a sort of appropriate question to ask, given the framework or the way we're supposed to interpret the maths. But I can't help wondering it, given that it strikes me that, you know, in order to interpret it as a penalty for complexity, you have to say, you have to assume that what you're starting with is simplicity. And I guess there would be real life situations in which you don't start with simple beliefs. And then presumably, this would tell you, well, if your beliefs are complex, then they ought to stay complex, because that penalty is a penalty for straying from that kind of complexity, which seems, you know, different from how it's usually described.

Yeah, Stephen S?

And would you say that that falls into our more recent move from science as creating models and fitting things to those models, like in the most simple way, to be in modelers, like with post-normal science and even Pristen's work with the COVID. So there's a belief in engaging with a complex non-resolvable system question, which is different to science's traditional role, which is to have an answer and to fit things to it.

Well, one of the most exciting and sort of scary aspects of active inference is the idea that it's applied, not just as a way of understanding what our target systems are doing. So not just as a way of understanding what a biological system or a brain that I'm studying is doing, but also as a way of understanding what I myself I'm doing when I represent that target system in order to try and understand it. You get a kind of loop in a way, in a sort of sense in which you feel as though you could model your own activity by using these technical tools which active inference is giving you.

But then you feel like how could that be done in a way that doesn't invite any kind of infinite regress?

That's why it's scary as well as exciting. It's exciting because it seems to potentially provide some kind of extra power or insight into our own activities as scientists or as modelers of any kind.

But it's scary because it seems to then drop you into a kind of spiraling hole of the modeler who is modeling themselves, modeling themselves, etc.

That's something which I can't think around even informally, let alone try and get to the formalism of it.

That was the very structure of paper number 36. So even in its qualitative application, just at the terms and concept level, it does turn the mirror

back and that is speaking to the pre and post or pre post trans distributional aspects of ActInf.

All the funny things that Dean says as well and the strange loop there.

So what that made me think of is like the statistics and then the famous quote like the statistician can come in for the postmortem or something like that.

But a lot happens before the statistics happen, like the design of the experiment.

And so the optimal experimentation and then the knowledge that it will be passed through a distributional filter in the big macro of the scientific method, like planning months in advance for how many minutes and hours one is going to count insects or how many microliters of a fluid they're going to move.

That is sort of the implementation phase. And then there's a design time. And so that's also what ties science to engineering in a way that it hasn't been

in design thinking in a lot of other areas too. Dean?

Yeah, this is awesome. So, I mean, one of the things that we talk about is some of those density questions around

what's the linear aspect of it and what's the solenoidal aspect of it. And we also talk about walking into a room of mirrors with a mirror suit on. I mean, it all comes together in the sense that there, I think,

one of the things that Steven Mann pointed out is that there are, I think there are, one of the things

I think we can say about active inference that's consistent is that it has at least a minimum of two

in that back and forth, up and down the density bump and around the density bump, you can go either

direction. And so that, that has me asking whether when every time we, we say that this is a framework,

where there were only giving it half its due, whether it's a framework and also a filtering exercise,

like, does there have to be a minimum of two, not just the boundary, but then what is actually going

on dynamically within that border so that we know when we're crossing, when we know when we're within

the boundary that we've maybe are looking at empirically, right? Because we selected our unit

of analysis. But then when we know when we've crossed outside that, when we've, when we've gotten to a

place where the mesh is holding certain things back, while other things that are of a smaller

circumference are still able to sort of pass through. And so I think as a, as a, as a guide,

that question of, as we model, as we model, the model is the minimum. I think if we just say the model,

we don't do it justice. And if we just look at that, as we model, the act of doing that, I think, again,

we're looking at sort of a side of beef and expecting to be able to milk it. And I don't think

that that works. I think you have to have the whole thing, or else you don't really have active inference.

So that's, I think that's kind of where we're at when we're coming to the place where we're trying

to decide what goes in, and what doesn't go into a guide. And it's a struggle, right? Like, there's no, there's no question, it's probably one of the hardest things that you can do, because the majority of people that are guides are there because they've been there before, versus the one who hasn't. And that would require upending that that basic premise, which is, no, everybody has to get into the boat, and we're sailing off onto the onto the event horizon. That would be the only way that we would change that. So the actual nature of guides would only change if everybody had to get their skin in the game, and start at that basic place of not knowing. But that will leave that for another, another, another live stream. But at this point, that's what's really interesting here. One of the assumptions that we made is, somebody has taken the time to go and search and explore, and come back with information, and the other people don't have it. And the only way that changes is if everybody gets in the boat, and sails off in hopes that they don't come to the edge of the cliff and fall off.

So...

Okay. A lot there. Thank you, Dean. Just a few other points. People have different perspectives on what is core or not. Some of the uncertainty, let's get at it with the ontology development, and making some definitions and translations, and the formalisms clear, and the history of science, and all of that. But even then, it might be pretty fundamental. Like, if it includes action and inference, is variational methods necessary? So, who knows? People will have different interpretations just about a lot of things. And so that's, again, all part of the fun. So, Steven?

Steven

Steven

So, one question I'd be interested in is, talking about guides in terms of that book, Life A User's Guide, that's a very deep... I read it for a long time, and then I lost it. But it's sort of like, it takes you into all these aspects of people's lives in that Parisian neighbourhood. And it never gives you an answer. It doesn't, but in some ways, it's leaving it to your intuition. It's leaving to your integration. And I'm wondering where you see that, that integration that's happening at a non-perspectival level. The integration, the intuition, the affective consciousness that is part of our knowing, and how that relates to this. And maybe if it's even something that's not part of science, but is part of this kind of work.

Steven

Well, I mean, firstly, you sort of cracked the code in a way, because the title of the paper was directly and explicitly stolen from the correct book.

Steven

It wasn't necessarily a sort of the best joke because it's not...

Steven

In the sense in which the book is about an incredibly hyper specific and detailed description of a system. you know an apartment block at a specific time point on a specific day of the year gave me the idea about okay it's like it it's not true that we ever really represent things that way it's not

true that science intends to represent things that way in the way that correct does and he's sort of making fun of but also he's he's incredibly constrained in the way that he wrote the book as i understand there's some there's some really crazy games that he plays with language in writing it and the course for example translates is a huge amount of difficulty in translating it because how do you know how do you translate the game that he's played in french into a game that you can play in english while writing the english version of the novel so definitely the idea of representation and sort of naturalism was on my mind but the sort of the short reason we called it that is because we wanted the punchy title we wanted something that people would think was accessible um but definitely when it came to casting about for punchy and accessible titles that was the one that primarily came to mind um where it comes to sort of affective consciousness and sort of experience again it's it's a little outside my wheelhouse i i was very very impressed upon by the novel even just in the english translation because i can't read french i was really really sort of moved by it in various different ways you know intellectually emotionally things like that there's a lot about sort of representation in there just the concept of representation uh and that's something that i'm you know interested in but again that pulls sort of in the opposite direction from an activism or affective consciousness and things and the very i'm very much situated in that kind of formal dry approach to philosophy the kind of an electric tradition where we try and chop things down we think we're scientists you know inspecting the human condition under a microscope um and so but you know i can't deny that in reading a novel it flows over me in the same way you know i'll experience it in a similar way that you experience it i hope

yes blue

so totally funny um i was also had a question about a guide but like on the opposite of like life a user's guide i was reminded of the movie beetlejuice um and they like the couple dies and they have this like handbook for the recently deceased right like the the core of like the theme that they keep well did you read the guide well did you read the guide and it's apparently like a very thick dense book like that no like you can't i mean we're dead we don't know what to do and we're freaking out we don't have time to sit down and read this book and so um i i was reminded of that just kind of totally juxtaposition but it's like when you are creating a guide it's very difficult to like make it usable right like the math versus the territory like what what will people read and be able to take away and walk away with some kind of general understanding versus like if you really are going to know what to do when you're dead in this situation you've got to read the book um you know how to like haunt the house and all these things and so like do do we really need to like pull grip of the fbp to really sit down with the guide you know whatever like three thousand page manual or like where how do you distill the essence of something that's so complicated like what is it to be dead or what is the fbp right it's hard that's such a good example like it's a handbook for the handbook for the long-lived the i guess the free energy principle as opposed to handbook for the recently deceased uh yeah what what is the the line that they have in that it's something like some incredibly dry statement which is absolutely no help at all like operational parameters vary from

manifestation to manifestation or something and yet i mean the idea behind it was that the philosophers that we talked to that urged us to try and research and write this were pointing out that when they attempted to read those papers in the active inference tradition even those that were billed as tutorials it was like trying to read the handbook for the recently deceased it was if not too long then at least too dense and they couldn't pick apart those statements because they were too condensed and so yeah the difficulty was un-condensing it pulling it apart selecting what was necessary and then presenting it in an accessible way i mean all i'll say is that it took sort of on the order of years to do it we we sort of got commissioned as it were in 2019 i started researching it in earnest at the beginning of 2020 and the first pre it went through a few you know drafts and people gave us a lot of comments behind the scenes but the first preprint came out in december 2021 so yeah it's for me at least and for us the authors it was that difficult to troll through the literature find what was necessary and then present it in a presentable way but that's such a great example thanks so much that hadn't crossed my mind but i love that film and the concept of the handbook as well so just a few comments on that it's not called free energy principle a user's guide it's actually very specific it's about the variational free energy and the expected free energy f and g respectively what's the fep so that's sort of where it takes you and it's very subtle then um that was pretty interesting a handbook for the long lived and it reminded me about metacognition in learning and about what we expect and prefer and think we can know and how like just being a um learner and like a teacher's assistant seeing how there'd be lecturers that would give um maybe it could be thought of in sort of an effort g framework of another day but they would simplify and learners would have high precision on a very simple model and then other times they would um convey too much of the complexity and so at the end of the quarter even if the students learned a ton and they would still feel like lost in a bigger space and there's like good and bad aspects of all of that so uh i think this is kind of like a snapshot it's kind of cool that it is so recent that was only a few months ago right now that it's february 22 but this is very like recent in and still kind of ongoing that you're looking to update it to so it's sort of like a kind of window where there's interactions with the community and hopefully people really think about it and um the people for whom a section isn't clear have an opportunity through this affordance we have now as scientific publishing with preprints like to ask or correct or you missed a citation here that is awesome and it's a world apart from sort of three people with it on their desk who are delaying it in review and then maybe they have good comments maybe they don't but here there can be something that's so much beyond that as this is just like one thing that you're working on and you're developing and uh then one last comment on that is like spm the textbooks are still kind of classics cult classics we might even say and it's not that the notation hasn't changed a lot because it has in some cases it's a snapshot and there's some amount of linking back to the past that's clarifying and then there can be an understanding that there's drift and focus and evolution of techniques and the spm textbook it's going to have sentences that you wouldn't include if you're writing it today great that's why it's different but also there's something about it that helps inform so every snapshot that's well intended always is good dean yeah i think one of the

other things that's really interesting about this is is stephen m is kind of pointed to the the length of time when the question was posed can you can you summarize this for us can you synopsise this for us basically what the philosophy it sounds to me is what the philosophers were asking was can you go and do the figuring out and then present that to us in a finding out way will you save us time and so i think what that's one of the things we have to keep front of mind when we look at this because those philosophers now didn't do the two years of heavy lifting and searching and foraging there's that's a whole different aspect that figuring out is a whole different processing method and mechanism than the finding out right and so again if you want people if you want people to come away from guide use with more than the finding out they're still going to have to get in the field and do the figuring out themselves and it's that interaction part which thank thank goodness for blue and beetlejuice because that was again that was just a wicked example right because it the timing of this the whenness of this really really matters and i know that in a lot of educational settings time is at the essence right 13 weeks x number of hours blah blah blah we tend to get that we tend to that's the only part of the whenness that we focus on not should the guide or the plan be the thing that initiates all the action or is that the thing that falls out afterwards if we're doing figuring out it has to fall out afterwards not before and we have a really hard time with that because traditionally it's always been the legitimizing of guides because they've been there before and they'll help you find out but they don't necessarily help you figure out just because the amount of time

yes please stephen m i mean yeah so both of the things that you both just said daniel and then dean like they're both open up huge topics which uh have so many interesting questions and so many interesting ways of approaching them um dean i i think i agree 100 with what you just said uh they've done the they're going to have to use our guide as a kind of shortcut in a way to doing the initial stuff the finding out but it behooves them in a way to do some extra work of figuring out afterwards the the thing that that most sort of resonates with me regarding is just the idea that our like mandate what we had to do in producing this was give them the ability to work through like the examples themselves because that's a part of i mean you know i'm not sure how you're distinguishing these two things but to me that's a part of figuring out as well as just finding out what to me we're not just saying this is a measure of inaccuracy and this is the algorithm by which you minimize it we're saying look at this example with these numbers you know in the imaginary situation and if you work through it and if you you know print out the graph yourself you will find that this is where the minimum lies for example and you will compare that with this other thing so we're doing the sort of the the first tiny step in helping them figure out how they would figure it out in a way if you see what i mean giving them the the the means to go ahead and do it again themselves in a different kind of situation but the sort of the other thing that it that it brings up i mean i shouldn't say that you know they they like michael weisberg didn't say you know go and do this for me go and be my like research assistant for two and a half years to print this thing uh he along with uh peter godfrey smith who is another person uh not connected with the journal in that capacity but very interested in having this framework explained for a wider audience pointed out that first and foremost we ourselves

would get some kind of like plaudits for having done the work having helped people to understand it so you know there was the the personal selfish incentive to do it um but also that it would be something that would benefit the community in a sort of in a positive way not something that we would be giving them for free as it were but something that we would be creating for everybody to draw on and use and just improving how the community approaches this question so in the same sense in which you know in in the best case academia is a wonderful collection of people who are all helping each other to you know find those shortcuts do the finding out without going through however long it takes to sift out the the stuff that is not necessary we're all helping each other to do that you know i can read somebody else's paper who is working for two years on something completely different and having split up that labor divided that labor up we can both sort of benefit from it um sorry i'm going on far too long going on no no no no one i'm only answering one question one last thing about the minimum of two things though if we could get you you and connor heinz into a live stream i think my mind would be blown just because that math and this philosophy in one proximal location because that that to triangulate off that would be amazing because that's essentially what you have to do you have to put the finding out and figuring out together i don't think again i don't think it's one or the other i think it's our ability to sort of be able to hold up both at once and appreciate how each of those things fills in the picture yeah and i do want to daniel you should tell me when i should uh stop or you know please just jump in and interrupt me anytime uh but you know i just wanted to sort of respond to your points as well um the other good thing since i'm in an optimistic mood about academia uh like you say the kind of pre-publication pre-print sending it out and enabling people to give us comments back on it if you think about it like is that metaphor with going to a classroom and teaching before you go in and teach you have to pare away all of the nonsense the stuff that's too complex get it down to the level that's simple enough so they can understand but has enough information so they can actually learn something in the really good case of getting a kind of course evaluation you do a whole course you get a bunch of students tell you what was right what was wrong what went too fast what went too slow you know and hopefully there's a kind of average where it averages out and you improve it sort of next time you make it easier for the learners because the learners have told you what was right and what was wrong about it it's always that kind of feedback process obviously in real life situations it doesn't always work that well getting that kind of feedback um but if you're sort of lucky enough to start off from a decent point as i think we have at this paper you can uh you can get to a kind of point which once it is snapshotted because that's the other important thing i think that this kind of pre-publication peer review is really useful but the that there shouldn't be constantly evolving documents all the time there should be some kind of snapshots once it's snapshotted it's like it's as good as it can be at that time even if it becomes redundant eventually even if it uses language which won't be used in 10 20 years once it becomes something that people can refer to valuably for some kind of certain amount of time afterwards that's really the best that you can hope for but you know certainly you know just a paper this is just a paper it's nothing uh more than that really just one thought on that like on the finding out and figuring out connecting that to the peer review

it's almost like we're all co-learners in participatory engaging education so we want to make it easier for the learners which is us and we're all co-learners so sometimes we find out that would be somebody with more domain expertise giving us the feedback on the paper and then in the side channel maybe there's another time and place where it's a different roles in that relationship but like and then there's the figuring out and then maybe there's other things here too that's actually um not like feedback from the bottom up but beginners questions abductive logic applications adversarial or contrarian even questions that is always like what is active inference why does f matter um what does the fvp have to do with active inference what does evolution have to do with natural selection what does natural selection have to do with drift if those were not resolved then they're going to be some open ones in this area as well and i think there is also another analogy with sort of like uh in a um concordance with evolutionary theory and like darwin's dangerous idea dennett in 95 that type of book um perhaps has already or soon will be written for something like fvp because it has that same ability to like spawn an adjective and then reintegrate the adjective back into the corpus so it was like there was like evolution of this type and then people add like a modality or a subdomain or a new word and then it rejoins the main thread so we have like deep affective sophisticated contrastive dot dot dot but it's going to reintegrate because it will be versioning and that returns to like blue's point about what is core what isn't and how will that change and so and then you just kind of close the loop with the snapshotting but the recognition of the process but we have to make the static modification in the niche but of course it's a process so it's really um interesting where we've gone today yes stephen m and then s oh yeah just very quickly um i think the point about contrarian questions is a very good one um it's often philosophers main method of figuring something out beginning with the contrarian question that is uppermost in their mind going out and trying to find the answer to it and certainly in this case um on on my side when i was doing my part of writing the paper um i was starting with what on earth is this maths and what is the interpretation of these different components of it um and that was just for the for the maths when you get onto the sort of philosophy side the interpretation of the dialectic i mean one thing i haven't said is that there are some quite skeptical portions of this paper you probably you know noticed when you read it and we tried to you know pull back a little bit but what we wanted to do was provide you all for figuring out and you always want to try and do that in good faith you know you always want to do it in a way that's not combative or dismissive you want to do it in a way that is uh designed to help you and the person that you're asking come to a consensus about the answer thanks steven okay steven s yeah i really find this helpful this gratuitous use of the term guide because you chose that title which maybe if you weren't doing a pre-print maybe people i noticed people a bit more creative with the titles which um then that's maybe also a nice benefit but by going there this discussion has been really helpful i'll be i often hear people when they talk about the findings from research that oh we're going to roll out a toolkit we'll do a toolkit and then when they talk about a guide it's really a sort of a guide a guide to teach the toolkit and but here the what in a way you're looking at especially if you're getting into the more uh deeper tacit knowledge is what's the guiding principles to being a modeler what's it about

being a modeler and that's where well these kind of conversations are useful but i think that the the type of stuff that's been brought up here are things where you know they're not resolved down to a toolkit level but it opens a question for someone to have a conversation about um so i just thought i'd mention that and uh what your thoughts are about that challenge of someone becoming a practitioner rather than necessarily an academic uh disseminator

yeah it's tough because i'm really not sure in part because all that i'm sort of trying to be right now is an academic disseminator so i would have to shift into a sort of different gear to think about what it would mean to be a practitioner um i mean i i would start by asking you what you take being a practitioner to be and then see if i can you know resonate with something in that well for me it's um it's more contextual and it's more about being able to adapt and um evolve based on a situation and often you find the practitioner work is when you're in a face-to-face performative context in a way where you have to perform either the dialogue i suppose in science when you're in the field of practice in other areas it could be the performance of giving a performance um be a performance as a as a as a teacher it could be a performer being an actor being a um a workshop facilitator being a therapist being uh whatever that might be um and i suppose that that that that question about how something being a guide a toolkit or some way to access um um the sort of heuristics and ways of knowing which aren't even representational in the kind of the way that we can easily externalize so i suppose there's some of that there's the stuff that we can't take a perspective on directly because we can't necessarily encode it into artifacts and place them in our environment in words however that knowing is a skillful practice that has been sort of cultivated so i i kind of feel there's there's a need for that and there's a there's a big move towards that now with um the world trying to embrace transdisciplinarity participation complexity where you you have no way to make any sort of movement forward without the ability to to find other ways to hold the meaning um dean dean just to just to sort of piggyback on stephen i think i think if you want to try to draw a comparison if you're a disseminator the result can often be something like a desire line a wearing in a following and if you're looking at the generative model you're looking at a densifying at a foraging and then a and an accumulation and in that in that respect It's quite clear what, from a practical sense, one's not better than the other, but one results in quite a different result than the other.

It's something to think about.

And then one other thought on this practitioner and scientist, though, I hope we see a hundred times more and more different ways to learn and apply active inference.

Just total first thought would be like when Steven was describing a practitioner, contextual and adapting on situations with really lower levels of control.

It's on a continuum, but like a crisis or a transition moment or an adversarial scenario has a lot of an implicit factor.

And it's not always as focused on like the coolest, calmest, most collected, stegmergic artifact modification. It's more like the firefighting was done appropriately in that scenario, given what it was.

And then the scientist generates niches like epistemic niche, laboratory niche, computer niche, that allows them to generate scenarios where they can apply distributions.

And then when they apply the distributional methods where there shouldn't be a distributional method, it's going to be, unfortunately, a silent error most of the time because of how little paid attention to all these differences are.

So, Steven M.

Steven M.

Yeah, this is really tricky to think about.

I mean, I think I agree that knowing ought to be considered a kind of skillful practice.

That was one thing that you said, Steven, that I really sort of jumped out at me.

Steven M.

In part, as being not sufficiently emphasized elsewhere, certainly not sufficiently emphasized in the tradition of philosophy that I come from.

There tends to be a fairly clear distinction between knowing in terms of representational knowledge and knowing how in terms of skill.

Steven M.

And there's not that often the consideration of the kinds of things we would usually consider representational knowledge as underpinning a kind of skillful practice.

But when I think about the way that I'm thinking about philosophy conferences and the kinds of conversations that go on, it's pretty clear to me that having a conversation with somebody else about a topic that is so sort of steeped in law and intellectual recognition and texts as having a philosophy conversation is what it requires is a kind of skill, a kind of a smooth,

They're kind of automatically picking out the topics as you go along, not sort of sitting back and thinking very hard on your own before coming back 30 minutes later and responding to the things that your interlocutor has just said.

It has all the hallmarks of a skill, as far as I can see.

And what happens is, well, the way that I have been brought up to view it is you do all your thinking at home or you do all your thinking in the armchair.

And then when you come out to have the conversation, you're ready with that skill.

Because you've done all of that slow, solitary work first, the kind of propositional or representational kind of work, then when you're ready, you come out and you practice discussing it.

Thanks. Great point.

It's a lot like the representational dimension is kind of the nonlinear knowledge model, which cannot be conveyed in a linear sequence.

Linear presentation in video and in writing is rhetoric.

And that was one of the most interesting areas in the paper was like the claims justification, the whole second part.

And I'm happy that we spent almost all the dot zero in our conversation leading up to it.

Because there's a whole paper and it sounds like there's so much more to explore.

But like the part about the rhetoric and the way that people not just have an edge in their knowledge graph, their armchair knowledge graph or their relational database, not just an edge between concepts, but actually when a question is raised to move to that answer first or the first topic that's mentioned.

Like if somebody asks, what is a markup blanket?

The first noun, what is it going to be?

Well, for a cell, or is it because of Bayesian or what is the actual first rhetorical node that somebody gives?

And those kinds of knowledge representations are really interesting too.

It's not even just that it's the most important thing because that's not how people always structure their speech.

So like there's both sides.

We need the non-linear part so that we can actually build a bigger structure than just an oral script.

But then it has to be in that like runtime that you kind of described at the philosophy conference.

It has to be presented in a linear way.

And if there weren't disagreements or like contrasts in people's perspective, then they would just be reading off the same book.

So maybe we'd love to hear your thoughts on any of these latter sections in four.

Like which one of these areas of claims, we won't stay long, but just what of the areas of claims or the relationships between areas was interesting to you or like you learned while reading about it?

Because you had the justificatory links between dialectical categories with mathematical, empirical, and general.

So maybe just one edge or one section of those.

What did you find interesting?

Oh, was that an open question or were you asking me?

I wasn't sure.

I guess to you.

I mean, I can definitely start with the first section, which was the mathematical to empirical direction.

And as you said, we for each edge of the triangle, we were only sort of considering one direction, mainly for considerations of space.

What's most interesting to me about this is just that it is an established research field called computational cognitive science.

When the specific maths that is being looked at is certain kinds of neural networks or other kinds of computational systems.

And the specific empirical system that is being looked at is a brain, especially the human brain.

And what we really wanted to emphasize in this section was that the kinds of tools that have already been developed by computational cognitive science.

So that includes the models and the explanatory tools and the justificatory or inferential tools could be applied to active inference models as well.

So this was our attempt to say to the philosophers, we already have some equipment we can use when what we're trying to do is evaluate active inference.

So whether we call it deflationary or not, there's a lot of relevance in emphasizing which parts are specific to

active inference and which ones aren't.

Like bringing action into the loop is not a novelty in the last 15 years.

Doing Bayesian statistics, doing variational inference.

And so that work helps us really highlight the parts that do have unique and different interpretations.

Like this is definitely one of my favorite paragraphs.

I think that we've come across and gotten the highlight on the discussions.

Like, and then you even brought it up that it's actually in the distillation.

There is a novel, not just implicit knowledge or notational connection, but a rhetorical element that arises.

And then also one with like educational and didactic implications.

So that's really interesting.

Dean?

Yeah.

And I think when I was reading a paper, Stephen, I noticed the number of times you had the word evaluate, evaluate, evaluate,

which again is another one of those, do you get, does this pass, pass muster, pass a standard, pass, what goes, what goes on and what gets held back?

What gets continued attention and what do we now ignore?

And that's why I like when you were talking earlier about, well, when we put a guide together, we just want to basically walk into the bar and make sure that the 98 things that we really can't pay attention to first, that doesn't pass.

What does pass are maybe these two first nascent ideas.

And so, again, you don't need to hear this, but if you want to just a factory link, that is one right there in real time in terms of what, what, what carries on, what can we build off of and what falls away?

Because it doesn't, it doesn't seem to pass whatever evaluate, evaluative methods we've implied.

But then again, that goes to when do we start evaluating from the, from the moment we started, do we want it to be something a little bit more creative and hold that thing open a little long before we jump into it?

We want to assess.

Okay.

Okay.

So in the last just few minutes.

Yeah.

Steven, you want to give a response there or would you like to have a little bit of a closing statement?

Let me give a quick response, which is just that I agree exactly with what you said, Dean.

And yeah, we wanted to show that some things could be evaluated very directly just by working through the numbers and that give some directions for how other things could be evaluated.

And you're right.

It's like passing.

It's sort of how a philosopher evaluates a scientific theory as being sort of in a sense, first level plausibility or first level acceptability, how it passes that it's, it's kind of a philosopher's technical term.

But I can sort of give a kind of closing statement that calls back to what you just mentioned, Daniel, that paragraph, which is just that the thing that I'm thinking about most often when I think about active inference right now

is the interpretation of P in the expected free energy function and the sense in which it can have a probabilistic interpretation as well as an interpretation of a kind of preference.

That is the thing which I really want to try and get to the bottom of.

Again, just by working with very, very simple models, try and throw some numbers around, try and intuitively grasp what that model can represent.

We have one in there with the example of the cat, but I want to try and expand it a little bit.

Yeah, I'm glad that you like that paragraph because it's the thing that is causing me the most.

It's the thing that's itching the most that I'm attempting to scratch right now.

All right.

So we'll each give a closing thought.

So just what did people enjoy?

What will they look forward to talking about next week?

So Stephen, select first.

Well, just thank you a lot for a really great engaging conversation and lots of insights.

I'm really interested in that idea of the P 's.

So I think that's very interesting.

And I'd be interested in how that maybe applies both on the sensory side and the action side.

So how much is the probabilities and preferences for sensing as much as or is it that one is more for sensing, one's more for action.

But I'm very interested in all of that.

So thank you very much.

And look forward to more conversations.

I'll go ahead.

So thanks for entertaining all my interrogations.

I appreciate it.

If you're going to come back next week, I would encourage you to check out A Tale of Two Densities because like on the tale of Stephen with the P 's and the Q 's.

I'd be curious to think about or to hear what your interpretation of that is and if you would maybe change or maybe you will.

I mean, it's just a preprint.

So maybe you have a versioning affordance to kind of consider that in your explanation in the beginning.

I'm curious or if you think it's one of these superfluous extra components that is for the handbook for the recently deceased, not necessarily, you know, like the 13 steps of dying or whatever.

So I'd be curious to hear what you have to say about that.

Dean, anything?

Yeah, it's just that I don't want it to drop down to the sort of quantum level, but I think that when you're doing a guide, it's hard not to sort of try and juggle both the dynamic aspects of what you're trying to share and the stable aspects of what you're trying to share.

So you want the screwdriver to remain robust and constant and then you want it to also be able to be manipulated in such a way that it drives the screw.

And there's a dynamic and a stability piece to this guiding.

It's really, really hard to, as you say, square.

I don't know that it can be squared.

I think they both have to be maintained.

And so my sense is, is that I, I didn't know what you were going to do in terms of the guiding, but I think the way that you laid it out did slow things down, did allow the teeth of the screwdriver to get set.

And then you could actually do something dynamic with it.

So really appreciate that.

And maybe, maybe next week we can talk about when the screwdriver is used for creative purposes other than driving screws.

Because I think what the math, if we use the, the geometry and the statistics and the philosophy all in one concoction, I think that's the basis or the minimum that we have then to be able to be, do creative things and not just recombinatorial things.

So maybe we can touch on that.

I would be really curious to hear how you see this going off in creative ways without going.

And because the big fear people have is that just means destabilization, where I'm saying it has to start from both stable and the potential to move.

Thanks, Dean.

Stephen, any last thoughts?

Just to thank you all very much for engaging with our paper, for inviting me on the stream, for asking me really great questions and raising great points that just had never crossed my mind.

This is exactly the reason why I was hoping to do this.

I am going to try and come back next week.

I can't make any promises right now, though, because next week is a bit of a heavy week for us.

But just, yeah, the prospect of going and reading that paper and then being able to respond in a more intelligent way to please question is making me really want to do it.

Not to mention all of the other topics, which, you know, we haven't talked about yet for the rest of you as well.

So I will let you know during the week.

But fingers crossed.

I'll be back again next week.

But thank you all.

Thank you.

Yeah, really appreciated everybody sharing their perspective.

And you're welcome back anytime.

So see you, everybody.

Peace.

Peace.

Thank you.